MIL-L-60244A(MU)
6 April 1970
SUPERSEDING
MIL-L-60244(MU)
17 September 1965

MILITARY SPECIFICATION

LINKAGE ASSEMBLY: 8287301

1. SCOPE

1.1 This specification covers the performance requirements, related quality assurance provisions, and packaging data for one type of linkage assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWING

U.S. Army, Frankford Arsenal

F8287301	Linkage Assembly: 8287301

PACKAGING DATA SHEET

8287301	Packaging of Linkage Assembly: 8287301

FSC 1240

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(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The linkage assembly shall be manufactured in accordance with Drawing F8287301 and drawing pertaining thereto.

3.2 General specification. - The requirements of MIL-F-13926 listed below shall be met wherever applicable.

a. Order of precedence.
b. Dimensions and tolerances.
c. Inorganic protective surface finishes.
d. Part identification and marking.

3.3 Workmanship.- The workmanship requirements of MIL-F-13926 shall be met. The exterior protective surface finishes shall be free of voids which may affect their protective qualities or appearance.

3.4 Extreme storage temperatures.- The linkage assembly shall show no evidence of physical failure due to storage at temperatures ranging from -80° to +160°F.

3.5 Extreme operating temperature.- The linkage assembly shall meet the applicable requirements of 3.7.1 while exposed to and thermally stabilized at -65° and +150°F.

3.6 Vibration. - The linkage assembly shall show no evidence of physical failure due to being vibrated successively along three (3) mutually perpendicular axes. Vibration shall be a simple harmonic motion, the frequency varying either linearly or logarithmically from 5 to 55 to 5 CPS in five minute cycles for a total of 15 minutes along each of the axes. The amplitude of vibration shall be maintained constant at 1/64 inch (1/32 inch total excursion).

3.7 Orientation.- The requirements of 3.7.1 and 3.7.2 shall be met with:

a. The assembly positioned as shown on Drawing F8287301.
b. The 2-1/4 inch x 2.499 inch mounting surface, that is located by the 0.250 inch dimension shown on Drawing F10542037, horizontal within 30 minutes of arc.
c. Tube 8635542 originally positioned 10 degrees 30 minutes ± 3.0 degrees with respect to the 2-1/4 inch mounting surface (89°49′35″ with respect to the angular position of Arm 10542031).
d. A constant resistance load of 3.0 inch pounds ± 1.0 inch pound applied to the oldham coupling.

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e. A suspended weight (free rotating) of 8 pounds ± 2.0 ounces positioned on the exposed end of Shaft 8287311.

3.7.1 Operability. - This requirement shall be met with the inner race of the Bearing 8570108, on the free end of the linkage secured against rotation. The direct force, applied horizontally, requiard to change the angular value of Surface "X" to 41 degrees, zero degrees, then back to 22 degrees 30 minutes shall not exceed 4.0 pounds. The direct force required while the linkage is thermally stabilized at -65° and +150°F shall not exceed 6.0 pounds.

3.7.2 Backlash. - Backlash between Link 10542030 and Coupling 8620152 shall not exceed 0.10 mil.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) linkage assemblies plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials. including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

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4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of product" provisions of MIL-STD-105.

4.3.2 Examination and tests.- Components and subassemblies shall he inspected in accordance with the inspection provisions contained in MIL-F-13926, and the Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP'S, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MlL-STD-105, and the sampling plans specified in Tables I, II and III herein. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL:	NONE		
MAJOR:	AQL, .65% Defective		
101.	Vibration	3.6	4.6.3
102.	Backlash	3.7.2	4.6.4.1
103.	Operability	3.7.1	4.6.5
104.	Workmanship	3.3	4.6.6
MINOR:	NONE		

NOTE: The tests of Table I shall be conducted at a temperature between +60° and +90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and rejection" as specified in MIL-STD-105.

4..4 Special sampling.

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4.4.1 General. - One linkage assembly shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II

TABLE II

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable drawings - visual
302.	General specification	3.2	MIL-F-13926 - visual

4.4.2 Environmental. - Three linkage assemblies shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall have met the requirements and tests in Table 111 and shall then meet the requirements and tests in Table I.

TABLE III

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Extreme storage temperature (-80° to _ 160°F)	3.4	4.6.1
304.	Operability of linkage assembly (-65° and +150°F)	3.5, 3.7.1	4.6.2, 4.6.5

4.4.3 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

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4.5.2.2 contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test methods and procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "'Examination" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Orientation and backlash.- The special testing equipment utilized to mount and orient the linkage assembly for the backlash test shall conform to the following requirements.

(a) The testing equipment shall be capable of mounting and orienting the linkage assembly as specified in 3.7.

(b) A mounting block shall be incorporated on the test equipment that shall permit mounting and securing the 2-1/4 inch x 2.499 inch mounting surfaces of support 10542037 in a horizontal plane within thirty minutes of arc.

(c) An adjustable and moveable holding device shall be a part of the test equipment that shall permit the mounting, securing, fine adjustment and movement of Connector 8635540. subsequent to the mounting and securing of Connector 8635540, the holding device shall be capable of adjustment so that the axis through Tube 8635542 can be positioned at 10 degrees 30 minutes ± 3.0 degrees with respect to the 2-1/4 inch x 2.499 inch mounting surfaces (85°49'35" with respect to the angular position of Arm 10542031).

(d) A viewing collimator, capable of measuring any line of sight deviations in elevation or depression, in seconds of arc, shall be a part of the test equipment. The viewing collimator shall be located and mounted in such a manner so that the viewing collimator may be elevated and 'depressed in direct relation with and on the same axis as Coupling 8620152.

(e) A front surface reflecting mirror shall also be provided on the test equipment. The mirror shall be located and mounted directly in front of the objective end of the viewing collimator, in alinement with the line of sight emitting from the viewing collimator. The mirror shall be capable of being adjusted in all planes.

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4.5.2.3.2 Operability. - The special testing equipment utilized to mount and orient the linkage assembly for the operability test shall conform to the following requirements.

(a) The testing equipment shall be capable of mounting and orienting the linkage assembly in its normal operating position with Surface "X" of Coupling 8620152 originally positioned as shown on Drawing F8287301.

(b) A device shall be incorporated on the test equipment that shall permit mounting and securing the 2-1/4 inch x 2.499 inch mounting surfaces of Support 10542037 in a horizontal plane within thirty minutes of arc.

(c) A direct force type tester with a range of from zero pounds to 7 pounds, the accuracy of which shall be ±0.5 pound throughout the entire zone of excursion, shall be provided as part of the test equipment.

(d) An adapter that is capable of positioning the Tube 8635542, as specified in 3.7c, shall be provided as part of the test equipment. The adapter shall provide for immobilizing the ball of the spherical bearing, on the free end of the linkage, against rotation during the test. The adapter shall also permit the outer member of the spherical bearing to rotate freely about the ball.

4.5.2.3.3 Vibrating machine.- The vibrating machine shall be capable of providing the amplitude and frequency specified in 3.6. The vibrating machine shall also be capable of vibrating the linkage assembly along 3 mutually perpendicular axes while the linkage is maintained in the position specified in 3.7.

4.5.2.3.4 Vibration adapter.- The adapter shall be capable of positioning and maintaining the linkage assembly, on the vibrating machine as specified in 3.7a, 3.7b, and 3.7c.

4.5.2.3.4 Set-up and calibration.- Rigidity, stability, ease of maintenance, and adaptability of set-up and calibration equipment with complete provisions are also prime requirements. All set-up and calibrating equipment necessary for certification of the testing equipment accuracy shall be furnished and will become part of the testing equipment.

4.6 Test methods and procedures.

4.6.1 Extreme storage temperatures.- Testing equipment utilized in this test shall be in accordance with the "Test facilities" requirements of MIL-F-13926 and the conditions of 3.4. The linkage assembly shall be exposed to each specified temperature for a minimum of five hours. Subsequent to the temperature exposures, 'the linkage assembly shall be stabilized in an ambient temperature between 60 degrees and 90 degrees Fahrenheit and subjected to the visual, tactile, and audible examination of 4.6.3. The

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linkage assembly shall show no evidence of physical failure as specified in 3.4. The linkage assembly shall then be subjected to the tests specified in 4.6.3 through 4.6.6 inclusive, to determine overall compliance with 3.4,

4.6.2 Operability, extreme temperatures. - Testing equipment utilized in this test shall be in accordance with the "Test Facilities'! requirements of MIL-F-13926 and the applicable conditions of 3.5. The linkage assembly shall be exposed to each specified temperature for a period of five hours. At the end of each five hour period and while still at the temperature extreme the linkage assemblies shall be subjected to the test specified in 4.6.5 and shall conform to the applicable requirement of 3.7.1 for each temperature extreme.

4.6.3 Vibration. - Perform this test with the linkage assembly positioned on testing equipment that conform to 4.5.2.3.3 and 4.5.2.3.4. The linkage assembly shall be vibrated in accordance with Procedure IX, Part 1, of MIL-STD-81O, except that the frequency, frequency cycling, and amplitude of vibration shall be as specified in 3.6. Upon completion of the vibration cycling the linkage assembly shall be subjected to the tests specified in 4.6.3.1.

4.6.3.1 Examination. - Subsequent to the test in 4.6.3 the linkage assembly shall be thoroughly examined for any evidence of physical failure. A visual and tactile examination shall be made for missing or damaged parts. A tactile examination shall also be conducted to determine looseness or improper functioning of moving parts. An audible examination shall be made of the linkage assembly by shaking it and listening for loose internal parts or broken components. Any one of the possible defects defined above shall be cause to consider an assembly defective. The linkage assembly shall be subjected to the tests in 4.6.4 through 4.6.6 inclusive, to determine overall compliance with 3.6.

4.6.4 Orientation. - The following test shall be performed with the linkage assembly positioned and conditioned in accordance with 3.7a through 3.7e on special testing equipment that conforms to the requirements of 4.5.2.3.1.

4.6.4.1 Backlash. - With backlash removed for direction of travel, actuate the Connector 8635540 by means of the testing equipment until Surface "X" of Coupling 8650152 is positioned 22 degrees 30 minutes as shown on Drawing F8287301. Direct a line of sight through the viewing collimator to the reflecting (auto-collimating) surface of the mirror. While viewing through the collimator, adjuct the mirror until the horizontal reticle line of the reflected image is superimposed on the horizontal reticle line of the viewing collimator. Record the position of the commenctor. Continue actuation of the connector in the same direction of travel until Surface "X" of the coupling is positioned, approximately 33 degrees, then actuate the connector in the opposite direction of travel (with backlash included) until the connector is returned to the recorded lateral position without overtravel. Record the boresight point (intersecting point of the

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horizontal and vertical reticle lines) of the reflected image, as measured on the viewing collimator. Any vertical displacement of the boresight point of the reflected image is the amount of backlash. Backlash shall not exceed the limit specified in 3.7.2.

NOTE 3: During this operation (4.6.4.1), if overtravel occurs the test shall be nullified and the procedure specified for backlash shall again be performed.

4.6.4 Operability. - Special testing equipment conforming to the requirements of 4.5,2.3.2 shall be utilized for this test. Position and condition the linkage assembly as specified in 3.7a through 3.7e, and attach the testing equipment to measure direct force, push or pull to the spherical bearing on the free end of the linkage. Direct force shall be measured while actuating the connector, by means of the direct force tester, in the appropriate direction until Surface "X" of Coupling 8620152 is rotated through its travel range specified in 3.7.1. The direct force applied at the connector to rotate the coupling shall conform to the applicable requirement of 3.7.1 at the temperature specified in 3.4.

4.6.6 Workmanship. - The linkage assembly shall meet the requirements specified in 3.3 when subjected to a visual and tactile examination.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 8287301. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The linkage assembly is a component of the gunner's fire control system in the Armored Reconnaissance Airborne Assault Vehicle XM551. This assembly serves as a link between the periscope input coupling and the weapon and provides for elevation of the line of sight coaxially with the weapon.

6.2 Ordering data.- Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable packaging data sheet number (see 5.1).
(d) Provisions for first article testing (see 4.2).

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1240-A217

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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